

Euclid's Elements — Book I

Proposition 40

Formal Reconstruction — Technical Documentation

Jurek Róžański
Military University of Technology, Warsaw, Poland
jerzy.rozanski@wat.edu.pl

CGJTEAMLAB

Equal triangles which are on equal bases and on the same side are also in the same parallels.

1 Introduction

Proposition I.40 is the equal-base companion to Proposition I.39. The four triangle propositions at the end of the first area block therefore form a natural pattern:

- I.37 : same base + same parallels $\longrightarrow$ equal content,
- I.38 : equal bases + same parallels $\longrightarrow$ equal content,
- I.39 : same base + equal content $\longrightarrow$ same parallels,
- I.40 : equal bases + equal content $\longrightarrow$ same parallels.

The present Lean theorem keeps the same synthetic content language used since I.35. Equality of triangles is represented by

```
HilbertScissorsEquicomplementable
```

rather than by a numerical area function. Its conclusion is geometric: the two top vertices lie on a Hilbert incidence line disjoint from the common base carrier.

The main architectural fact is that I.40 does not need a second converse-area argument. The difficult same-base converse has already been isolated in I.39. The only additional task is to reduce the equal-base configuration to a same-base one. For that reduction the formal proof reuses the congruent-copy construction developed for I.38.

This reuse was not merely cosmetic. The first draft of I.40 repeated the angle construction, segment layoff, SAS argument, oriented parallel argument, and parallelogram criterion already present in I.38. Inspection of I.40 showed that two I.38 helpers had been stated with unnecessary proposition-specific data. Their signatures were generalized, the full project was rebuilt successfully, and I.40 now calls that common construction directly. Thus the proposition records a genuine API improvement discovered by a later proof.

No new content principle is introduced in I.40. The theorem depends on I.39, and therefore inherits the positivity assumption already isolated there, but I.40 itself adds no new axiom, no numerical area, and no new polygon semantics.

2 The Classical Configuration

Euclid’s statement concerns two equal triangles on equal bases lying in one straight line and on the same side of that line. In the current formal statement the triangles are

$$ABC \quad \text{and} \quad DEF,$$

with equal bases

$$BC \cong EF.$$

The four base points are placed in the strict order

$$B - C - E - F,$$

encoded by

$$\text{Between}(B, C, E), \quad \text{Between}(C, E, F).$$

The points B and C are given on an incidence line **base**. The betweenness hypotheses then force E and F onto that same carrier. The top vertices A and D satisfy

$$\text{SameSide}(A, D; \mathbf{base}).$$

This is a deliberate specialization of the fully general Euclidean statement. Wilkins’ Study Note points out that a complete treatment of arbitrary equal bases on one line requires six positional configurations. These arise from the possible relative positions of the two base segments together with the position of the second top vertex relative to the auxiliary parallel through the first. Euclid explicitly discusses only one of those cases.

The present Lean theorem does *not* claim to formalize all six configurations. It works with the ordered separated-base arrangement $B - C - E - F$, the same carrier normalization already used by Proposition I.38. The advantage is that the equal-base geometry can be reused without reopening a large order case split. A future fully order-free wrapper could normalize other base arrangements before calling this theorem; that normalization is not part of the current proposition.

Figure 1 shows the actual formal configuration and the two conceptual reductions used in Lean.

3 Classical Proof

In Euclid’s displayed configuration the two equal bases share an endpoint: the triangles are ABC and CDE , with bases BC and CE . Join the two top vertices A and D . The aim is to prove that AD is parallel to the base line.

Assume that AD is not the required parallel. Through A draw another line parallel to the base. Let it meet the side of the second triangle, or its extension, at an auxiliary point F , and join F to the appropriate base endpoint. Proposition I.38 applies to the original triangle and the auxiliary triangle because their bases are equal and they lie in the same parallels. Hence the auxiliary triangle is equal to the first triangle. By hypothesis the first triangle is equal to the

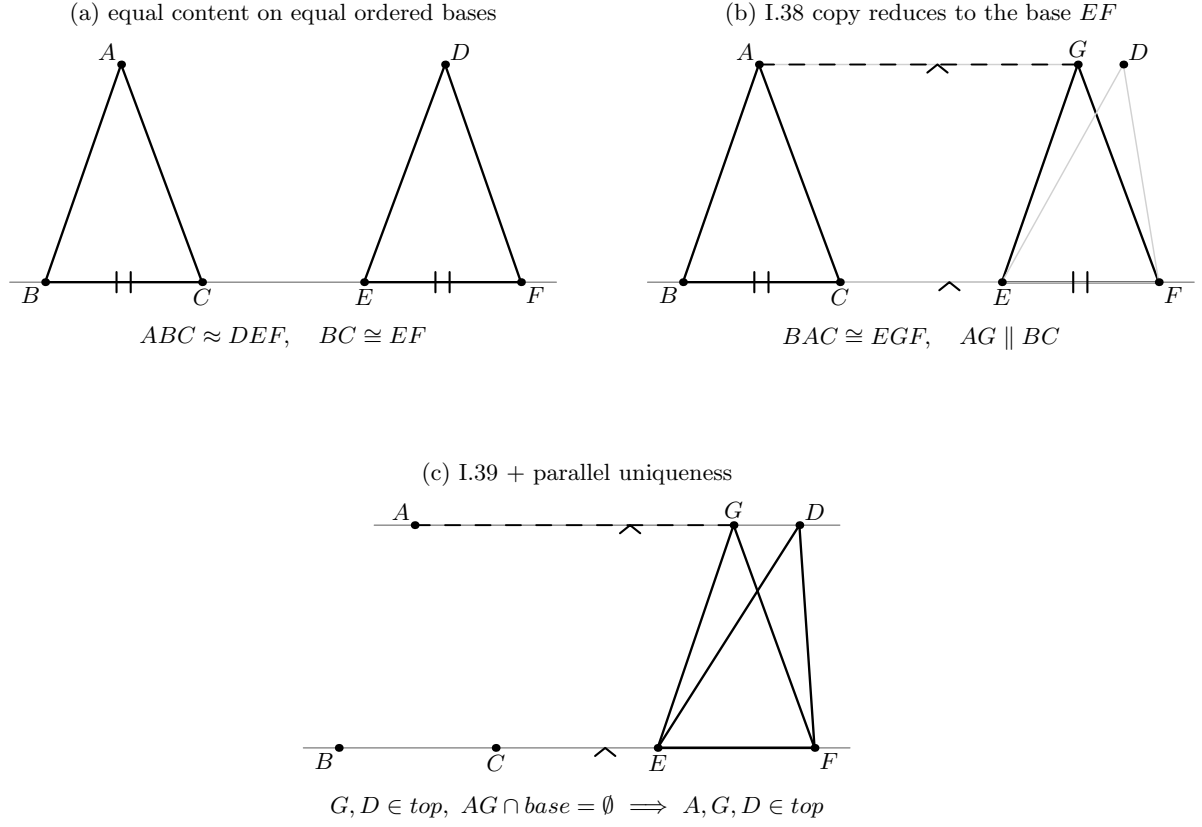


Figure 1: The formal reconstruction of I.40. (a) The given triangles ABC and DEF have equal content, equal ordered bases BC and EF , and top vertices on the same side of the common base carrier. (b) The reusable I.38 construction produces a point G such that $BAC \cong EGF$ and $AG \parallel BC$; internally this comes from the parallelogram $AGEB$. Hence ABC and GEF are related by the stronger scissors equality, and transitivity gives equal content for GEF and DEF . (c) Proposition I.39 applied on the common base EF places G and D on a line top disjoint from base . The carrier of AG is also disjoint from base ; since both candidate top lines pass through G , Euclidean parallel uniqueness identifies them and places A on top as well.

second, so transitivity of equality forces the auxiliary triangle to be equal to the second triangle. But one of these is geometrically a proper part of the other. This is the “greater equals the less” contradiction used by Euclid.

The classical dependency in the displayed case is therefore

equal triangles on equal bases
 $\Downarrow$
 draw a parallel through the first top vertex
 $\Downarrow$
 auxiliary triangle in the same parallels
 $\Downarrow$ I.38
 auxiliary triangle = first triangle
 $\Downarrow$ C.N.1
 auxiliary triangle = second triangle
 $\Downarrow$
 proper-part contradiction
 $\Downarrow$
 the top vertices lie on the required parallel.

The historical proof and the current Lean proof therefore share the idea that I.40 should be reduced to an earlier equal-base theorem, but they reduce in opposite directions. Euclid uses the forward theorem I.38 inside a reductio. The Lean development first uses the I.38 *construction* to move to a common base and then invokes the already completed converse I.39.

Wilkins’ expanded analysis is important here. Once the two bases are allowed to be adjacent, separated, or interlocking, and once both possible positions of the second top vertex relative to the auxiliary parallel are considered, six cases must be ruled out. The present formal theorem avoids claiming this full classical case analysis: its explicit $B - C - E - F$ hypotheses select one ordered base arrangement from the outset.

4 Equal Bases by Reduction to the Same-Base Converse

The central structural observation is that the passage from I.39 to I.40 is the converse analogue of the passage from I.37 to I.38.

For the forward direction, I.38 constructs a congruent copy of one triangle on the other equal base and then compares the copied triangle with the target by I.37. For the converse direction, I.40 uses the same copy construction, but now the copied triangle and the target triangle are equal in content and share *the same base*. Proposition I.39 can therefore be applied directly.

Schematically,

$$\begin{array}{ccc} \text{I.37} & \xrightarrow{\text{equal-base copy}} & \text{I.38} \\ \Updownarrow & & \Updownarrow \\ \text{I.39} & \xrightarrow{\text{same equal-base copy}} & \text{I.40} \end{array}$$

where the vertical symbols indicate only the forward/converse relationship, not a formal biconditional theorem in Lean.

This observation led to a small but significant refactor of `Proposition38.lean`. The helpers

```
i38_equal_base_copy
i38_copy_parallelogram
```

had originally accepted the second top vertex D and a hypothesis of the form $AD \parallel BC$. Inspection showed that these data were not intrinsic to the copy itself. What the construction actually needs is:

- one explicit base carrier;
- B, C, E, F on that carrier;
- the ordered-base data $B - C - E - F$;
- $BC \cong EF$;
- the point A off the base.

After the signatures were generalized, I.38 continued to build cleanly and I.40 could call the construction without repeating its proof.

The resulting Lean DAG is

$$\begin{array}{c}
ABC \approx DEF, \quad BC \cong EF, \quad B - C - E - F, \quad \text{SameSide}(A, D; \mathbf{base}) \\
\Downarrow \\
E, F \in \mathbf{base} \\
\Downarrow \quad \text{i38_copy_parallelogram} \\
\exists G : \quad BAC \cong EGF, \quad \text{SameSide}(A, G; \mathbf{base}), \quad AG \parallel BC \\
\Downarrow \\
ABC \sim GEF \\
\Downarrow \quad \text{content symmetry/transitivity} \\
GEF \approx DEF \\
\Downarrow \quad \text{I.39 on base } EF \\
G, D \in \mathbf{top}, \quad \mathbf{top} \cap \mathbf{base} = \emptyset \\
\Downarrow \quad AG \parallel BC \\
\text{carrier}(AG) \cap \mathbf{base} = \emptyset \\
\Downarrow \quad \text{Euclidean parallel uniqueness through } G \\
A, G, D \in \mathbf{top} \\
\Downarrow \\
\text{euclid_proposition_40.}
\end{array}$$

Here $\sim$ denotes the stronger relation

```
HilbertScissorsEq
```

while $\approx$ denotes

```
HilbertScissorsEquicomplementtable
```

5 Proof Architecture of the Reconstruction

5.1 Step 1: putting the second base on the common carrier

The first proposition-local helper is

```
i40_base_endpoints
```

It starts from $B, C \in \mathbf{base}$ together with

$$B - C - E, \quad C - E - F,$$

and proves

$$E, F \in \mathbf{base}.$$

The proof is pure incidence/order transport. Betweenness supplies both collinearity and endpoint distinctness, and line uniqueness transports the new points to the already chosen carrier.

This helper is intentionally local. It packages the particular ordered four-point configuration used by I.40; it does not introduce a new general notion of collinear base segments.

5.2 Step 2: reusing the I.38 congruent-copy construction

The second helper is

```
i40_copy_parallel
```

Its output is exactly the geometric bridge needed by the converse proof:

$$\exists G : \text{SameSide}(A, G; \mathbf{base}), \quad BAC \cong EGF, \quad AG \parallel BC.$$

The helper does not rebuild this geometry. It calls

```
i38_copy_parallelogram
```

which returns the stronger package

$$\text{SameSide}(A, G; \mathbf{base}), \quad AGE B \text{ is a parallelogram}, \quad BAC \cong EGF.$$

From the parallelogram one gets $AG \parallel EB$. Since B, C, E are collinear, parallelism is transported from the carrier EB to BC . The result is $AG \parallel BC$.

This is the point at which the I.38 refactor pays off. The substantial angle, ray, SAS, extension, and I.28 machinery remains proved once, in I.38. I.40 uses only its mathematical output.

5.3 Step 3: transporting equal content to the copied triangle

Triangle congruence gives the strong scissors relation

$$BAC \sim EGF.$$

The scissors representation of a triangle is insensitive to exchanging its first two vertices, so the proof rewrites this as

$$ABC \sim GEF.$$

The relation `HilbertScissorsEq` implies `HilbertScissorsEquicomplementable`, hence

$$ABC \approx GEF.$$

By symmetry and transitivity with the hypothesis

$$ABC \approx DEF,$$

we obtain

$$GEF \approx DEF.$$

No subtraction, cancellation, strict content comparison, or numerical area is used in this stage. It is ordinary equivalence transport in the existing scissors layer.

5.4 Step 4: reducing to I.39 on the common base EF

The copy construction puts G on the same side of **base** as A , while the public hypothesis puts D on that side as well. Symmetry and transitivity of **HilbertSameSide** therefore yield

$$\text{SameSide}(G, D; \text{base}).$$

The strict betweenness $C - E - F$ gives $E \neq F$. All hypotheses of I.39 are now available for the triangles GEF and DEF on the same base EF .

Proposition I.39 returns a line **top** such that

$$G, D \in \text{top}, \quad \text{HilbertLinesDisjoint}(\text{top}, \text{base}).$$

This is the only converse-content step in I.40. Consequently all strictness needed to turn equal content back into geometry is inherited from I.39 rather than reintroduced locally.

5.5 Step 5: identifying the final **top** line

It remains to place A on the line returned by I.39. The helper `i40_copy_parallel` already supplied $AG \parallel BC$. Let **topAG** be the incidence line through A and G . Since BC lies on **base**, point-line parallelism implies that **topAG** and **base** are disjoint.

Now both **topAG** and **top**

- pass through G , and
- are disjoint from **base**.

The Euclidean uniqueness of a parallel through an external point therefore forces

$$\text{topAG} = \text{top}.$$

Since $A \in \text{topAG}$, it follows that $A \in \text{top}$. Together with $D \in \text{top}$ and disjointness from the base, this is exactly the public conclusion.

6 Lean Formalization

The current file imports the two propositions that carry its geometry and its converse-content step:

```
import CGJteamLab.Proposition38
import CGJteamLab.Proposition39
```

The public theorem is:

```
theorem euclid_proposition_40
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D E F : Geo.Point)
  (base : Geo.Line)
```

```

(hBCE : Geo.Between B C E)
(hCEF : Geo.Between C E F)
(hBbase : HilbertIncidence.OnLine B base)
(hCbase : HilbertIncidence.OnLine C base)
(hBC_EF : Geo.Congruent B C E F)
(hSame : HilbertSameSide Geo A D base)
(hEqual :
  HilbertScissorsEquicomplementable
  Geo
  (hilbertScissorsTriangle Geo A B C)
  (hilbertScissorsTriangle Geo D E F)) :
Exists fun top : Geo.Line =>
  HilbertIncidence.OnLine A top /\
  HilbertIncidence.OnLine D top /\
  HilbertLinesDisjoint Geo top base := by
...

```

The ellipsis is documentation notation only. The repository theorem is fully proved. Unicode logical symbols from the Lean source have been normalized to ASCII forms in the listing above.

The two proposition-local helpers are:

```

i40_base_endpoints
i40_copy_parallel

```

The second of these calls the generalized I.38 helper

```

i38_copy_parallelogram

```

rather than repeating its proof.

The scissors part of the final theorem uses the following existing operations:

```

scissors_congruent
scissors_triangle_swap12
equicomplementable_of_scissorsEq
equicomplementable_symm
equicomplementable_trans

```

The line-placement part then uses:

```

euclid_proposition_39
HilbertPlaneIncidence.line_through
hilbert_mem_pointLine_iff_onLine
HilbertEuclideanPlane.parallel_unique

```


7 Relationship with Earlier Propositions

7.1 Proposition I.38

I.38 supplies the reusable equal-base geometry. Its original formal proof had to construct G by angle layoff and segment layoff, establish triangle congruence by SAS, recover the appropriate orientation through extensions and I.28, and package the result as the parallelogram $AGEB$.

I.40 revealed that this construction was more general than its original signature. The helpers `i38_equal_base_copy` and `i38_copy_parallelogram` were therefore generalized to accept the base carrier and off-line data directly. The public I.38 theorem continues to use them, and I.40 now reuses the same package. This is a concrete example of a later proposition improving the reusable boundary of an earlier proof.

7.2 Proposition I.39

I.39 is the decisive converse theorem. It says that equal-content triangles on one common base and on the same side lie between the same parallels. Its proof is where the strict content issue appears: the proper-part contradiction ultimately depends on the positivity principle isolated in the scissors layer.

I.40 does not reproduce that argument. Once the copied triangle GEF has been made equal in content to DEF , I.39 is applied directly on the base EF . Thus the foundational debt of I.40 is inherited from I.39 rather than multiplied.

7.3 Proposition I.37

I.37 is not called directly by `Proposition40.lean`, but it remains in the transitive dependency graph through I.39. It is the forward same-base theorem used inside the I.39 reductio. The four propositions I.37–I.40 therefore form a coherent block rather than four independent area arguments.

8 Hilbert and Book Zero Components Used

The proposition sits on several already established layers.

8.1 Incidence and order

Strict betweenness provides endpoint distinctness and collinearity. The helper

```
hilbert_collinear_on_line
```

transports E and F onto the chosen base carrier. The theorem also uses incidence lines through distinct points and the equivalence between membership in a point-line carrier and incidence on a chosen Hilbert line.

8.2 Side-of-line geometry

The public statement exposes

```
HilbertSameSide Geo A D base
```

explicitly. The copied point G comes with a second `SameSide` relation from I.38. Symmetry and transitivity then produce the exact `SameSide` hypothesis required by I.39. This information is invisible in the short classical phrase “on the same side” but is a first-class proof object in Lean.

8.3 Parallelism and parallelograms

The I.38 construction returns the parallelogram $AGEB$. The existing parallel transport lemmas convert $AG \parallel EB$ into $AG \parallel BC$ using the collinearity of B, C, E . At the final stage the Euclidean uniqueness of the parallel through G identifies the carrier through A, G with the line through G, D returned by I.39.

Thus Group IV enters through the Euclidean parallel layer. No continuity axiom is used by the proposition.

8.4 Scissors content

The content calculation is intentionally small. Triangle congruence gives a strong scissors equality; representation transport reorders the triangle vertices; and the existing equivalence operations convert and compose the relations. No new scissors constructor or theorem is introduced in I.40.

The strict content principle needed by the converse direction is already encapsulated upstream in I.39 through

```
hilbert_scissors_triangle_positive
```

I.40 introduces no second positivity or cancellation assumption.

9 Source Audit

9.1 Euclid and Joyce

The classical Proposition I.40 is explicitly the equal-base converse of I.38. In Euclid’s displayed proof, a parallel is drawn through one top vertex; I.38 makes the corresponding auxiliary triangle equal to the first triangle, and transitivity of equality then produces the impossible equality of a greater triangle with a lesser one. Joyce also notes that the transmitted statement is less precise than it should be: the bases must lie on one straight line, and the proof should not depend on their sharing an endpoint.

9.2 Wilkins

Wilkins makes the hidden configuration problem explicit. A complete proof for two arbitrary equal bases on one line must exclude six configurations; only one appears explicitly in Euclid’s proof. This is an important control on the formal statement. The current Lean theorem does not silently claim those six cases: it assumes the concrete order $B - C - E - F$.

9.3 Hilbert and Hartshorne

Hilbert’s area theory and Hartshorne’s Section 22 remain the controlling sources for the meaning of equal content. The present reconstruction keeps `HilbertScissorsEquicomplementable` distinct from numerical area and from unconditional finite equidecomposability.

For the converse direction, I.39 has already isolated the proper-part strictness issue. I.40 therefore needs no new area semantics. Its genuinely new work is only the geometric reduction from equal bases to one common base.

9.4 Borsuk–Szmielew and Greenberg

Borsuk–Szmielew remain useful for the incidence, order, side-of-line, congruence, and parallelogram infrastructure, but they are not used as the primary source for equal-content semantics. Greenberg similarly provides a control on the Euclidean/neutral boundary and distinguishes equal content from numerical area, while referring the detailed content theory elsewhere.

9.5 Beeson

The Beeson proof corpus is retained as a configuration audit rather than as the architectural source of the present proof. No Tarski-style proof trace is copied into I.40. In particular, the final Lean architecture is determined by the already developed Hilbert/I.38/I.39 interfaces.

10 Formalization Notes

10.1 Geometry

The geometric data actually used by the theorem are:

- the strict order $B - C - E - F$;
- one common base carrier;
- equal base segments $BC \cong EF$;
- SameSide data for A and D ;
- the reusable congruent-copy/parallelogram package from I.38;
- Euclidean uniqueness of the parallel through a point.

The theorem does not infer all possible relative positions of equal base segments. Those alternatives are outside its current statement.

10.2 Content

The public content hypothesis is exactly `HilbertScissorsEquicomplementable`. The local copy step uses the stronger `HilbertScissorsEq` because congruent triangles can be replaced directly. The proof then weakens this relation and uses only symmetry and transitivity.

There is no local subtraction, cancellation, comparison relation, numerical area, or whole-greater-than-part argument in I.40. The only strict content principle in the dependency graph is the positivity axiom already used to prove I.39.

10.3 Representation

The congruence theorem returned by I.38 is oriented as

$$BAC \cong EGF,$$

while the content theorem is stated for ABC and GEF . The formal proof therefore uses

```
scissors_triangle_swap12
```

twice. This is representational transport, not additional geometry.

Likewise, the conclusion of I.39 is expressed by an actual incidence line `top`, whereas $AG \parallel BC$ is initially expressed by point-line carriers. The final proof explicitly bridges these two representations before applying parallel uniqueness.

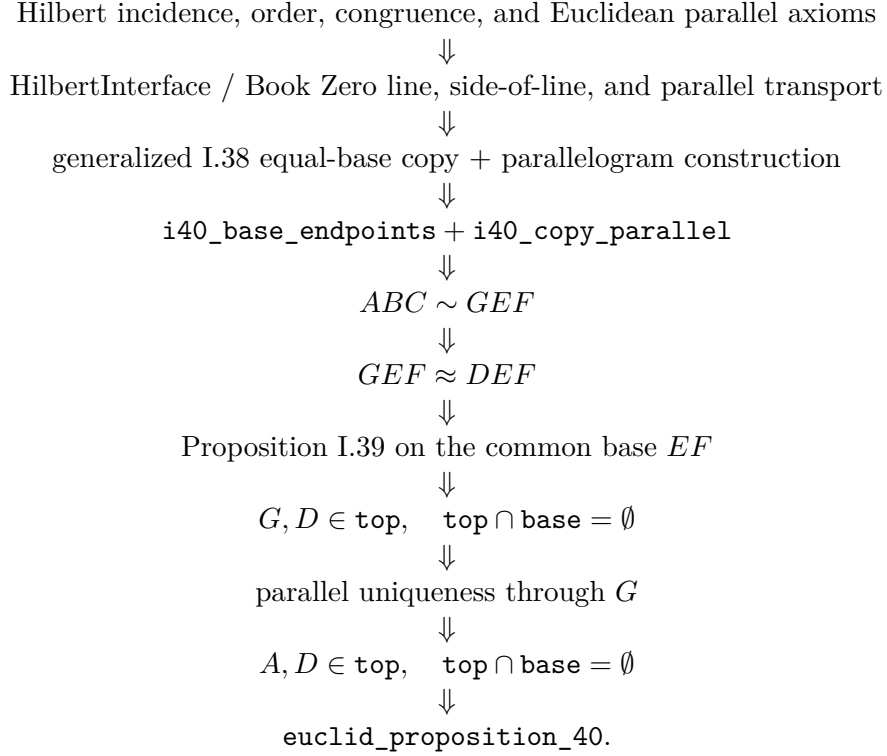
10.4 Axiomatic status

The public theorem assumes `HilbertEuclideanPlane`; it is therefore an Euclidean theorem in the current hierarchy. The final carrier identification uses the uniqueness of the parallel through G , and the reused I.38 construction also lives in the Euclidean layer. No continuity assumption is introduced.

On the content side, I.40 adds no axiom. It imports I.39 and therefore inherits the previously isolated scissors positivity assumption. This inheritance should not be confused with a new I.40-specific foundational principle.

11 Dependency Summary

The actual formal dependency chain is



The historical and formal DAGs are therefore related but distinctly organized:

Euclid : I.31 $\rightarrow$ I.38 $\rightarrow$ proper-part contradiction $\rightarrow$ same parallels,
 Lean : I.38 copy geometry $\rightarrow$ content transport $\rightarrow$ I.39 $\rightarrow$ parallel uniqueness.

New geometric axiom in I.40:	no
New content axiom in I.40:	no
Inherited content axiom:	yes — scissors positivity through I.39
New proposition-local helpers:	yes — two
New reusable declaration in I.40:	no
Reusable refactor required:	yes — generalized I.38 copy helpers
New Interface/Scissors theorem:	no
New Book Zero theorem:	no
New numerical area or polygon semantics:	no
Public theorem compiles:	yes
Full lake build:	clean
sorry in Proposition40.lean:	no

12 Conclusion

Proposition I.40 is a useful test of whether the preceding area block has produced reusable mathematics rather than proposition-specific code. Once I.39 supplies the same-base converse and I.38 supplies an equal-base congruent copy, the new theorem does not need another proper-part

proof. It transports the original equal-content hypothesis to a copied triangle on the base EF , applies I.39, and finishes by uniqueness of the parallel through the copied top vertex.

The most important development result occurred just before the final proof was written. I.40 exposed that the equal-base copy construction in I.38 had been stated too narrowly. Generalizing the existing I.38 helpers eliminated a large duplicate proof while leaving I.38 itself unchanged. The clean rebuild of the entire project confirmed that the refactor preserved the earlier proposition and produced a genuinely reusable boundary for I.40.

The content architecture also remains stable. No numerical area, new scissors operation, or new strictness axiom is introduced. I.40 inherits the single positivity debt already made explicit by I.39 and otherwise consists of geometry, representation transport, equivalence transport, and Euclidean parallel uniqueness.

Finally, the formal statement remains intentionally narrower than the most general classical proposition: it treats the explicit ordered carrier $B - C - E - F$. Wilkins' six-configuration analysis therefore remains a useful record of what a future order-normalizing wrapper would have to cover. The current theorem is complete for its stated configuration, builds cleanly, and fits the architecture established by I.35–I.39.